

JEE MAIN REVISION MASTER: MOTION IN A PLANE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Projectile Motion & Trajectory	8 PYQs	Projectiles
Standard	Relative Velocity in 2D (Rain-Man & River-Swimmer)	1 PYQs	Relative Motion

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Projectile Motion & Trajectory	$T = \frac{2u \sin \theta}{g}$, $H = \frac{u^2 \sin^2 \theta}{2g}$, $R = \frac{u^2 \sin 2\theta}{g}$, $y = x \tan \theta \left(1 - \frac{x}{R}\right)$
Relative Velocity in 2D (Rain-Man & River-Swimmer)	$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$, $t_{\text{min}} = \frac{d}{v_{\text{br}}}$, $\text{Drift} = (v_r - v_{\text{br}} \sin \theta) t$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Projectile Motion & Trajectory (High Yield — 8 PYQs)

Core Principle: 2D motion under constant gravity with independent horizontal and vertical components.

■ **Caution / JEE Pitfall:** Range is identical for complementary angles (θ and $90^\circ - \theta$), but flight time and maximum height differ.

■ **Speed Trick:** Equation of trajectory form $y = x \tan \theta (1 - x/R)$ allows fast calculation of range when trajectory coordinates are given.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Two bodies are projected with the same velocity. If one is projected at an angle of 30° and the other at 45° to the horizo..." **[Ans: B]**
- **[2025 Shift] Q#2 (MCQ):** "In the projectile motion of a particle on a level ground, which of the following remains constant with reference to time and position?..." **[Ans: N/A]**
- **[2022 Shift] Q#3 (MCQ):** "Two objects are projected at an angle θ° and $(90-\theta^\circ)$, to the horizontal with the same speed. The ratio of their..." **[Ans: N/A]**

3.2 Relative Velocity in 2D (Rain-Man & River-Swimmer) (Standard Yield — 1 PYQs)

Core Principle: Relative vector addition in planar motion, umbrella holding angle for rain, and river crossing optimization.

■ **Caution / JEE Pitfall:** To cross a river in shortest time, head perpendicular to the river flow ($\theta = 90^\circ$ relative to bank).

■ **Speed Trick:** To cross with zero drift (shortest path), swim upstream such that $\sin\theta = \frac{v_{\text{river}}}{v_{\text{swimmer}}}$ (requires $v_{\text{swimmer}} > v_{\text{river}}$).

Representative Source PYQs:

- **[2020 Shift] Q#4 (MCQ):** "Rain is falling vertically with a speed of 12ms^{-1} . A woman rides a bicycle with a speed of 12ms^{-1} in east to west direction. Wha..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Two bodies are projected with the same velocity. If one is projected at an angle of 30° and the other at 45° to the horizontal, then the ratio of maximum heights attained is

- (A) 3:1
- (B) 1:2 **[Correct]**
- (C) 4:1
- (D) 1:3

Correct Answer: B

Solution Analysis: For a projectile, the maximum height is given by $H = \frac{u^2 \sin^2 \theta}{2g}$. Here both bodies are projected with the same velocity, so u is the same for both. Also, g is the same. So maximum height depends only on $\sin^2 \theta$. For the first body, $\theta = 30^\circ$, $H_1 = \frac{u^2 \sin^2 30^\circ}{2g}$. For the second body, $\theta = 45^\circ$, $H_2 = \frac{u^2 \sin^2 45^\circ}{2g}$. So, $\frac{H_1}{H_2} = \frac{\sin^2 30^\circ}{\sin^2 45^\circ}$. We know, $\sin 30^\circ = \frac{1}{2}$ and $\sin 45^\circ = \frac{1}{\sqrt{2}}$. So, $\frac{H_1}{H_2} = \frac{(\frac{1}{2})^2}{(\frac{1}{\sqrt{2}})^2} = \frac{1/4}{1/2} = \frac{1}{2}$.

Benchmark Problem 2: KCET 2020 (Q#4)

Rain is falling vertically with a speed of 12ms^{-1} . A woman rides a bicycle with a speed of 12ms^{-1} in east to west direction. What is the direction in which she should hold her umbrella?

- (A) 30° , towards east
- (B) 45° , towards east
- (C) 30° , towards west
- (D) 45° , towards west

Correct Answer: None

Solution Analysis: The given situation is shown below. Here, velocity of rain, $v_r = 12\text{ms}^{-1}$ and velocity of woman, $v_w = 12\text{ms}^{-1}$. The relative velocity of rain w.r.t. woman is $v_{rw} = \sqrt{v_w^2 + v_r^2} = \sqrt{(12)^2 + (12)^2} = 12\sqrt{2}\text{ms}^{-1}$. The direction in which the woman should hold her umbrella is $\sin\theta = \frac{v_w}{v_{rw}} = \frac{12}{12\sqrt{2}} \Rightarrow \theta = \sin^{-1}(\frac{1}{\sqrt{2}}) = 45^\circ$, towards east.